

LAB # 7a

Construct 1D arrays in java

Objective

To Construct 1D arrays in java.

Lab Goals

To train on the construction of arrays in java and show how java treats arrays and the various elements of an arrays.

Introduction of Arrays

Java provides a data structure, the array, which stores a fixed-size sequential collection of elements of the same type. An array is used to store a collection of data, but it is often more useful to think of an array as a collection of variables of the same type.

```
dataType[] arrayRefVar;  
int [] myarray;
```

Defining Arrays:

An array in Java is a data structure that allows you to store multiple values of the same data type under a single variable name. To define an array, you specify the data type of the elements followed by square brackets [] and the variable name.

```
// Defining an array of integers  
int[] numbers;
```

Declaring Array Variables:

After defining an array, you can declare it by specifying the size of the array using the new keyword.

```
// Declaring an array of integers with size 5  
numbers = new int[5];
```

Example:

Following statement declares an array variable, myList, creates an array of 10 elements of double type and assigns its reference to myList –

```
double[] myList =newdouble[10];
```

Initializing Arrays:

You can initialize an array while declaring it or separately after declaration.

```
// Initializing array while declaring
int[] numbers = {1, 2, 3, 4, 5};

// Initializing array separately after declaration
numbers[0] = 1;
numbers[1] = 2;
numbers[2] = 3;
numbers[3] = 4;
numbers[4] = 5;
```

Accessing Arrays

You can access individual elements of an array using their index.

```
System.out.println(numbers[0]); // Output: 1
```

Reusing Array Variables

You can reuse array variables by assigning a new array to them.

```
numbers = new int[]{6, 7, 8, 9, 10};
```

Array Length

You can get the length of an array using the length property.

```
System.out.println(numbers.length); // Output: 5
```

1D Array

A one-dimensional array is an array that stores elements in a linear sequence.

```
int[] myVar = {1, 2, 3, 4, 5};
```

Example#1

Here is a complete example showing how to create, initialize, and process arrays.

```
public class Main{
    public static void main(String[] args){
        double[] myList = {1.9, 2.9, 3.4, 3.5};
        // Print all the array elements
        for(int i = 0; i < myList.length; i++){
            System.out.println(myList[i] + " ");
        }
        // Summing all elements
        double total = 0;
        for(int i = 0; i < myList.length; i++){
            total += myList[i];
        }
        System.out.println("Total is " + total);
        // Finding the largest element
        double max = myList[0];
        for(int i = 1; i < myList.length; i++){
            if(myList[i] > max) max = myList[i];
        }
        System.out.println("Max is " + max);
    }
}
```

Output

```
1.9
2.9
3.4
3.5
Total is 11.7
Max is 3.5
```

Example#2 Searching for an Element in an Array

```
public class ArraySearch {
    public static void main(String[] args) {
        // Initializing a 1D array
        int[] numbers = {1, 2, 3, 4, 5};
        int target = 3;

        // Searching for the target element
        boolean found = false;
        for (int i = 0; i < numbers.length; i++) {
```

```
        if (numbers[i] == target) {
            found = true;
            break;
        }
    }
    // Printing whether the element was found or not
    if (found) {
        System.out.println("Element " + target + " found in the array.");
    }
    else {
        System.out.println("Element " + target + " not found in the array.");
    }
}
}
```

Output

```
Element 3 found in the array.
```

Example#3 Copying an Array

```
public class ArrayCopy {
    public static void main(String[] args) {
        // Initializing a 1D array
        int[] numbers = {1, 2, 3, 4, 5};

        // Copying the array
        int[] copy = new int[numbers.length];
        for (int i = 0; i < numbers.length; i++) {
            copy[i] = numbers[i];
        }
        // Printing the copied array
        for (int i = 0; i < copy.length; i++) {
            System.out.print(copy[i] + " ");
        }
    }
}
```

Output

```
1 2 3 4 5
```

Exercise

- A. Create an array of String variables and initialize the array with the names of the months from January to December. Create an array containing 12 random decimal values between 0.0 and 100.0. Display the names of each month along with the corresponding decimal value. Calculate and display the average of the 12 values.

Expected Output:

```
January   : 45.67
February  : 78.23
March     : 12.89
April     : 56.45
May       : 89.12
June      : 34.56
July      : 67.89
August    : 23.45
September : 90.12
October   : 10.34
November  : 55.67
December  : 76.78

Average of the values: 52.34
```

- B. Write a Java program to calculate the average value of array elements.
- C. Write a Java program to reverse an array of integer values.
- D. Write a Java program that:
- Takes a string as input.
 - Splits the string into individual characters and stores them in an array.
 - Checks how many vowels (a, e, i, o, u, case-insensitive) are present in the string.
 - Counts the occurrences of each vowel separately.
 - Displays the results.

Input you have given from user: "Programming is Fun"

Expected output:

```
Characters: [P, r, o, g, r, a, m, m, i, n, g, , i, s, , F, u, n]
Total vowels: 5
Vowel counts:
a: 1
e: 0
i: 2
o: 1
u: 1
```

- E. Write a Java program to sort a numeric array and a string array.

Expected Output:

```
Original numeric array: [5, 2, 9, 1, 5, 6]
```

```
Sorted numeric array: [1, 2, 5, 5, 6, 9]
```

```
Original string array: [banana, apple, orange, grape]
```

```
Sorted string array: [apple, banana, grape, orange]
```